Reducing Energy Consumption of HVAC Systems Through Occupancy Optimized VAV Control

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Introduction

- Heating, Ventilation, and Cooling (HVAC) systems are responsible for 37% of the energy consumption of a typical university or educational institution building.
- HVAC systems being operated on inefficient schedules (i.e. running when spaces are unoccupied) waste thermal energy from heating/cooling and electricity, largely from fan loads.
- Many buildings at UCSD operate on a generous HVAC schedule of 6:30AM-10:30PM and do not rely on occupancy data to inform operations.
- **Variable Air Volume (VAV)** boxes allow for programmable control for air flow and conditioning within large buildings.

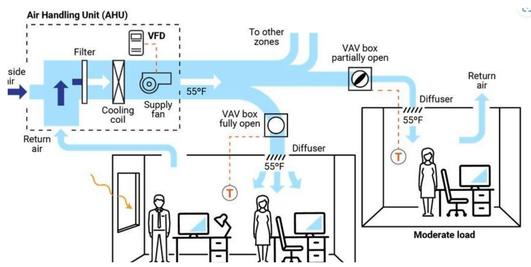

Figure 1: VAV Diagram

Proposed Solution

Implement a control method in 4 escalating levels which gradually tightens the HVAC schedule in order to better link operations to true occupied hours.

- By adjusting metrics such as the airflow and temperature band setpoints through set modes of occupancy, we hope to find savings for facilities management.
- **Higher levels of control** reflect a greater potential for savings with more difficult implementation.
- **Lower levels of control** maintain a steady baseline for that has the lowest potential for issues in implementation.

Objectives of Solution:

- Adherence to CA title 24 building code and ASHRAE guidelines.
- Maintain occupant comfort (CO_2, temperature, airflow).
- Increased energy savings from fan load and heating/cooling.

Control Levels: Each n level follows the n-1 level's rules.

Level 1	Mark room as Unoccupied mode for all hours of weekend, override if occupancy is detected.
Level 2	Tightened weekday Occupied mode hours. Ex: 6:30 to 8:30 vs 10:30. Full override to normal.
Level 3	Implemented time out clock to return to unoccupied after no motion is detected.
Level 4	Implement standby mode with direct link to occupancy sensors throughout weekdays. Ex: VAV 22

Methodology

- Building Automation Software (BAS) **METASYS** collects building information, such as light motion sensor activity and HVAC activity. This data is analyzed through visualization tool Grafana.
- We clearly define control modes for occupancy for the VAV boxes:

Occupied	Damper position of ~40% open, Temp Max 70°F, Temp Min 65°F
Unoccupied	Damper position of 0% open, Temp Max 90°F, Temp Min 55°F
Standby	Maintains intermediary temperature setpoints (not yet implemented)

- As we escalate, control methods will be tested and METASYS data will provide us with the opportunity to validate the effectiveness and any changes to air comfort.
- An important assumption is accurate occupancy data as the basis for our strategies.

Results & Discussion

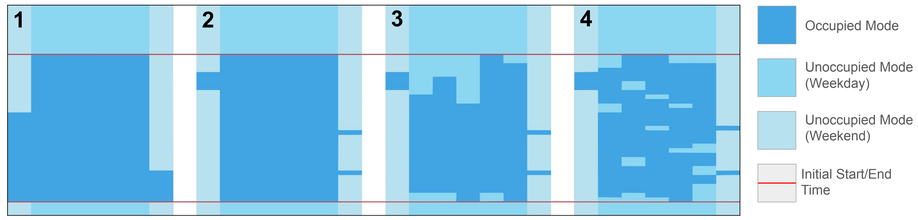

Figure 2: Visualizations of Levels 1, 2, 3, 4 of HVAC control.

Figure 3: kW consumption of AHU 1-7 in Center Hall over the period 12/04/24-12/22/24.

- During evening hours from 10:30pm to 6:30am the AHU are set to unoccupied mode and VAVs are closed, on weekends only some of the AHU are set to unoccupied.
- **Level 1:** In an idealized projection, where all weekend hours are unoccupied, 453.85 kWh are saved. (Savings of $59/day for electric costs of $0.26/kWh).
- **Level 2:** If 2 hours each week day, 77.2 kWh are saved, ($20.07/day savings). For a total calendar year of weekends being unoccupied and a 2 hour reduction of occupied hours during the day, this results in a 16.38% decrease in electric consumption by the AHU.
- **Levels 3 and 4:** Implement intermediary modes and timeouts that are unpredictable, thus we consider ideal schedules to get a baseline for savings.
- Note that these **AHU calculations only consider the reduced AHU fan load from changing the VAV damper positions.** While harder to quantify, the thermal energy savings from the widened heating and cooling setpoints are just as significant if not more.

Case Study: Center Hall VAV 22

- Center Hall 3rd floor is ideal for testing due to pre-existing lighting sensors, air handler submetering data access / control via Metasys, and zone use type (office / classrooms).
- **VAV 22 is tied directly to room occupancy**, significantly reducing the amount of time it spends open (effectively level 3 control). Despite this VAV 22's temperature control was effectively equal to all other VAV operations.

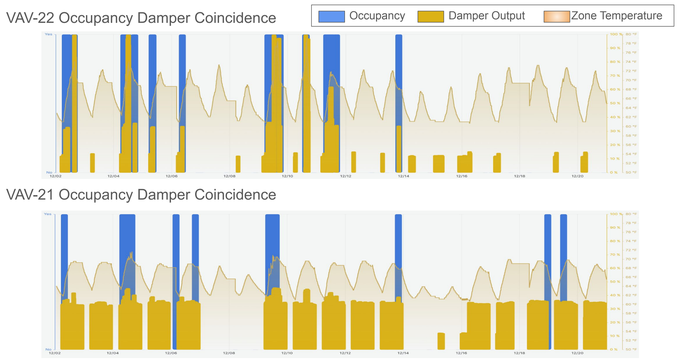

Figure 4: Room temperature, occupancy, and VAV damper position of VAV 21 and 22 from 12/02/24-12/20/24

Conclusion & Future Work

- VAV 21 & 22 case study fulfills all objectives of proposed solution, and demonstrates that tighter VAV control is possible.
- AHU fan load savings of roughly $59 per weekend day and $20 per weekday can be achieved through levels 1 and 2 of control.
- Additional savings come from reduced heating and cooling need.
- Shows the potential for good occupancy data to provide energy savings, supporting the investment into occupancy estimation.

Future Work:

- Pitch full plan to Facilities Management (FM) to gain permission to implement and test control schemes.
- Continuously monitor and assess METASYS data findings for impactful results and areas of improvement.
- Launch occupant comfort surveys in affected zones for additional validation and feedback.

References

Figure 1: https://www.pnnl.gov/projects/om-best-practices/variable-air-volume-systems
Figures 3 & 4: http://derconnect-brick.sdsc.edu:3000/d/fe12ed1b-1a5f-48d6-bd2f-4c8004c378e3/mae199-center-hall-academic-senate?orgId=1&from=1732089600000&to=1736409599000

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